

# evidence

## BOB-066 Evidence — Cross-Layer BOBA\_UPSTREAM\_PROXY Audit

**Revision:** 1 **Last modified:** 2026-08-15T12:15:00Z **Item:** BOB-066 — Lava P3: BOBA\_UPSTREAM\_PROXY in download-proxy + qBitTorrent-go + Jackett + compose env-forward **Status:** In progress (L1/L2/L4-partial WIRED; L3 residual gap) **Verifier:** challenges/scripts/upstream\_proxy\_wired\_challenge.sh **Run id:** 20260815T115831Z

### §11.4.108 Runtime signature — three-mode verifier evidence

Anti-bluff coverage per §11.4.115 polarity switch + §11.4.107(10) self-validation. Each mode's exit code is captured; every mode required to exit 0 for closure eligibility.

#### Mode 1 — --self-validate (analyzer-honesty against a synthetic golden-BAD tree)

The verifier runs its OWN detector chain against a temp tree where every wire is DELIBERATELY ABSENT (stub main.py without apply\_proxy\_env, stub search.py with zero trust\_env, stub httpx/proxy.go with no Configure/NewTransport/Proxy, empty docker-compose.yml). PASS = every detector correctly returned FAIL on the bad tree → the analyzer is not a bluff.

```
=== self-validate: run detectors against a synthetic golden-BAD tree ===  
PASS: self-validate – every detector correctly FAILED on the golden-BAD tree
```

**Exit: 0** — analyzer honesty confirmed.

## **Mode 2 — GREEN (RED\_MODE=0, default) — real tree**

```
=== BOB-066 four-layer cross-layer verifier ===
RED_MODE=0 (0=GREEN guard, 1=reproduce pre-fix defect state)

--- L1: download-proxy (Python) ---
[PASS] L1 config/proxy.py exports apply_proxy_env +
aiohttp_session_kwargs
[PASS] L1 apply_proxy_env() invoked at boot in download-proxy/
src/main.py
[PASS] L1 aiohttp ClientSession sites carry trust_env (9 uses of
_tracker_session_kwargs)

--- L2: qBitTorrent-go ---
[PASS] L2 internal/httpx/proxy.go exports Configure +
NewTransport + Proxy honoring BOBA_UPSTREAM_PROXY
[PASS] L2 httpx.Configure(cfg.UpstreamProxy) called in cmd/
qbittorrent-proxy/main.go
[PASS] L2 httpx.NewTransport() used in internal/api (torrent-
download client)

--- L3: Jackett ---
[SKIP] L3 Jackett proxy honor: reason=extension_absent (neither
compose env-forward nor ServerConfig.ProxyUrl wired) – tracked as
BOB-066 residual

--- L4: docker-compose.yml env-forward ---
[PASS] L4 docker-compose.yml service 'download-proxy' env-
forwards proxy vars
[PASS] L4 docker-compose.yml service 'qbittorrent-proxy-go' env-
forwards proxy vars
[PASS] L4 docker-compose.yml service 'qbittorrent' env-forwards
proxy vars
[SKIP] L4 'jackett' env-forward absent (see L3 SKIP; tracked)
[SKIP] L4 'boba-jackett' env-forward absent (not tracker-bound:
talks to Jackett loopback + local DB only)

=====
Summary: PASS=9 FAIL=0 SKIP=3
Notes:
- L3 residual: Jackett has neither env-forward nor
ServerConfig.ProxyUrl wired – see BOB-066 description
=====
PASS: upstream_proxy_wired_challenge – every wired layer verified;
residual gaps honestly SKIP'd

Exit: 0 — 9 PASS / 0 FAIL / 3 honest SKIP (§11.4.3 / §11.4.69
extension_absent).
```

### Mode 3 — RED (RED\_MODE=1) — §11.4.115 polarity flip

Under RED each presence-detector's polarity flips: PRESENT code becomes FAIL (because the pre-fix defect state RED simulates would have it ABSENT). On a tree where the fix has landed, EXPECT many FAILs — that IS the signal the polarity switch + detectors work end-to-end (they would have caught the defect if the code were un-wired). fail == 0 under RED would mean detectors ignore polarity OR the tree was reverted — both bluffs surfaced here.

```
=====
Summary: PASS=0  FAIL=9  SKIP=3
=====
RED-MODE VERIFY: 9 presence-detectors correctly negated under RED
- polarity switch works, RED→GREEN flip proven
```

**Exit: 0** — 9 detectors correctly negated → RED→GREEN flip proven.

### §11.4.108 four-layer audit — per-layer runtime signatures

| Layer | Sub-requirement                                                                    | Runtime signature                                                                                                                                                                                                                                                                    | Status       |
|-------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| L1    | Python download-proxy honors BOBA_UPSTREAM_PROXY (urllib subprocs + aiohttp)       | download-proxy/src/main.py:97 calls apply_proxy_env(); 9 aiohttp.ClientSession sites in merge_service/search.py carry _tracker_session_kwargs()=trust_env; config/proxy.py exports the API + UPSTREAM_PROXY_ENV="BOBA_UPSTREAM_PROXY" sentinel                                       | WIRED        |
| L2    | qBitTorrent-go http.Transport.Proxy honors BOBA_UPSTREAM_PROXY (or standard chain) | internal/httpx/proxy.go exports Configure/NewTransport/Proxy; cmd/qbittorrent-proxy/main.go:29 calls httpx.Configure(cfg.UpstreamProxy) at boot; internal/api/download.go:65 installs httpx.NewTransport() on the torrent-download client; socks5:// supported natively (remote DNS) | WIRED        |
| L3    | Jackett — env-forward OR ServerConfig.ProxyUrl                                     | Neither path present. jackett container has no proxy env-forward; no Go code touches ProxyUrl/ProxyType/ServerConfig in qBitTorrent-go/internal/.                                                                                                                                    | RESIDUAL GAP |

| Layer | Sub-requirement                                                            | Runtime signature                                                                                                                                                                                                                                         | Status                                       |
|-------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| L4    | docker-compose.yml<br>env-forward for every<br>tracker-touching<br>service | download-proxy, qbittorrent-proxy-go,<br>qbittorrent all env-forward<br>BOBA_UPSTREAM_PROXY + HTTP_PROXY +<br>HTTPS_PROXY + NO_PROXY (with default<br>bypass hosts). jackedt + boba-jackedt do<br>NOT (boba-jackedt is loopback-only so not<br>required). | 3/5<br><b>WIRED</b><br>(Jackedt<br>gap = L3) |

## Recommended follow-up for L3

File a scoped ticket (e.g. BOB-066b) implementing ONE of: - **(a)** Add proxy env-forward to the jackedt service block in docker-compose.yml (mirror the block at lines 15-20 of the qbittorrent service). Simplest fix. - **(b)** Add qBitTorrent-go/internal/jackedtconfig/proxy.go that on Jackedt-config load reads BOBA\_UPSTREAM\_PROXY and PATCHes Jackedt's ServerConfig.json ProxyType/ProxyUrl fields (Jackedt will honor them for its own indexer HTTP egress). Requires a boba-jackedt boot-time step + safe file-write discipline.

## Status decision (§11.4.6 / §11.4.108 / §11.4.226)

BOB-066 is NOT closed as Completed because L3 remains un-wired: closing with 3/4 sub-requirements met would be a §11.4.226 evidence-class-at-closure violation (the reported defect layer includes L3, and a source-green closure of an un-wired layer is exactly the pattern the anchor forbids). Bumped Queued → In progress; item\_history entry #65 records the audit with this evidence path; description carries the AUDIT NOTE 2026-08-15 block naming L1/L2/L4 wired + L3 residual + follow-up recommendation.